



ST BENEDICT'S

SUBJECT	Physical Science Paper 2	DATE	28 November 2019
GRADE	11	MARKS	200
EXAMINER	Mr McCoy	MODERATOR	Mr du Toit
NAME		DURATION	180 minutes
CLASS			

**LOW
ORDER**

55%

HIGH ORDER

45%

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 16 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages (i–ii). Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. Start each question on a new page.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Where appropriate express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	8	9	Total
Marks	20	25	17	27	18	33	23	17	20	200

QUESTION 1- MULTIPLE CHOICE

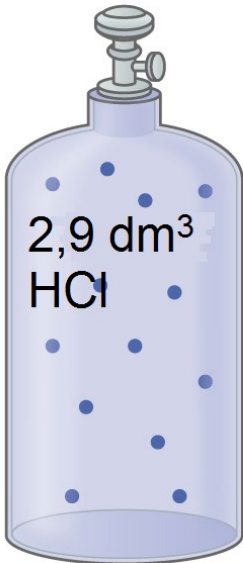
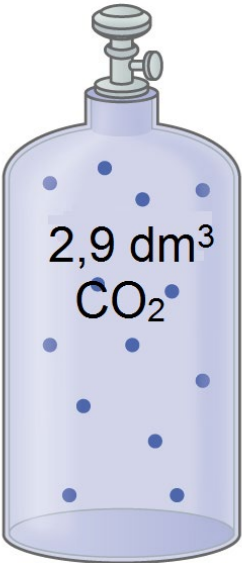
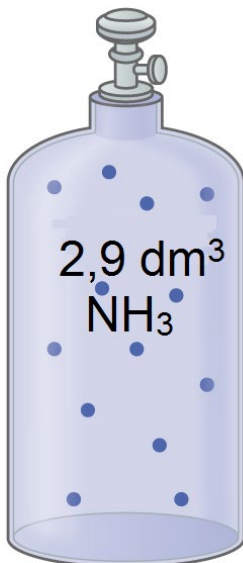
20 MARKS

Answer the following multiple-choice questions on your separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

- 1.1 Which one of the following statements represents the best explanation for the term 'electronegativity'?
- A. A measure of the tendency of an atom to attract a bonding pair of electrons.
 - B. A measure of the tendency of an atom to attract an electron.
 - C. A measure of the strength of a covalent bond.
 - D. A measure of the strength of an ionic bond. (2)
- 1.2 Which substance that has a giant covalent network structure.
- A. $\text{SiO}_2(\text{s})$
 - B. $\text{NH}_3(\text{g})$
 - C. $\text{C}_6\text{H}_{12}(\text{l})$
 - D. $\text{NH}_4\text{Cl}(\text{aq})$ (2)
- 1.3 Which substance is a **compound** with pure covalent bonds.
- A. $\text{SiO}_2(\text{s})$
 - B. $\text{PH}_3(\text{g})$
 - C. $\text{I}_2(\text{s})$
 - D. $\text{C}_6\text{H}_{12}(\text{l})$ (2)
- 1.4 Which substance exists as individual atoms, which interact through London forces only.
- A. $\text{Kr}(\text{g})$
 - B. $\text{Pt}(\text{s})$
 - C. $\text{I}_2(\text{s})$
 - D. $\text{C}_6\text{H}_{12}(\text{l})$ (2)
- 1.5 What mass of impure zinc metal having a purity of 89,5% is required to produce 0,044 moles of hydrogen gas at STP according to the reaction:
- $$\text{Zn} + \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$$
- A. 0,31 g
 - B. 2,85 g
 - C. 3,18 g
 - D. 6,43 g (2)

1.6 Each of the flasks below contain $2,9 \text{ dm}^3$ of gas at S.T.P. Which flask has the greatest number of moles?

(2)

 2,9 dm³ HCl	 2,9 dm³ CO₂	 2,9 dm³ NH₃	Same number of moles.
A.	B.	C.	D.

1.7 What is the number of atom in 6,4 moles of sulphuric acid, H_2SO_4 ?

- A. $3,85 \times 10^{24}$
- B. $4,21 \times 10^{24}$
- C. $1,16 \times 10^{25}$
- D. $2,70 \times 10^{25}$

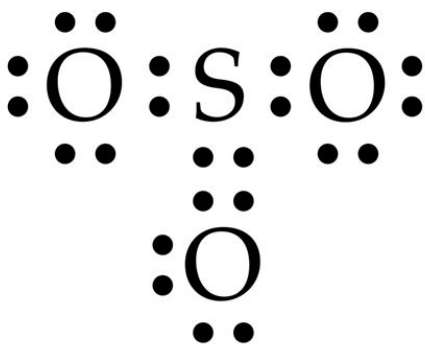
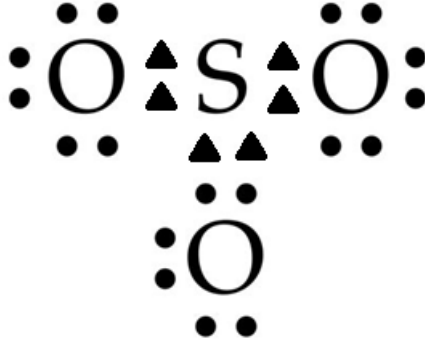
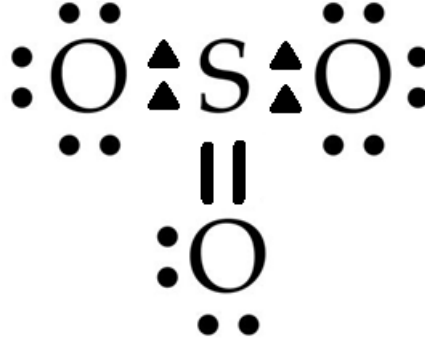
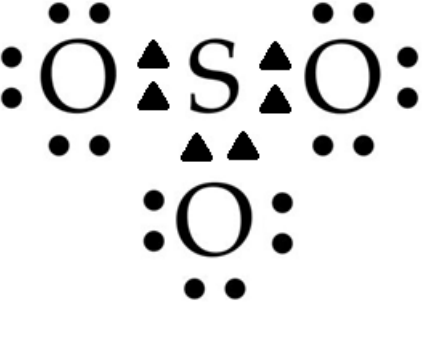
(2)

1.8 Which one of the following statements represents the best explanation for the term “heat-of-reaction”.

- A. Reactions which transform chemical potential energy into thermal energy.
- B. The net change of kinetic energy of the system.
- C. Reactions which transform thermal energy into chemical potential energy
- D. The net change of chemical potential energy of the system.

(2)

1.9 Which of the following is the most correct Lewis diagram for sulphur trioxide containing a dative covalent bond? (2)

A	B
	
C	D
	

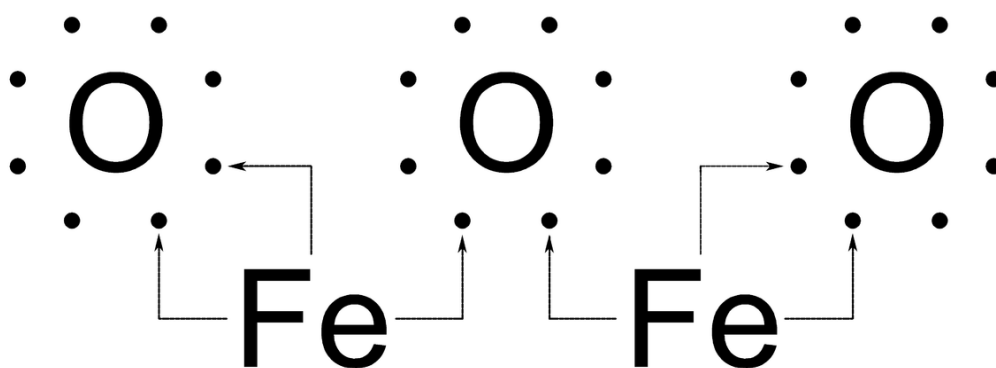
1.10 Which one of the following reactions will NOT take place with an alkene?

- A Hydrohalogenation
- B Combustion
- C Hydrogenation
- D Dehydration

(2)

QUESTION 2**24 MARKS**

2.1 Study the diagram below depicting the formation of an ionic compound.



2.1.1 Define *oxidation*. (2)

2.1.2 Name the atom being oxidized. (1)

2.1.3 Write the net ionic equation for the formation of the compound above. (3)

2.1.4 Refer to the diagram above.

(a) Draw a Lewis diagram of each ion that forms. (4)

(b) What will happen between these two ions to form the compound? (2)

2.2 Hex-1-ene is a non-polar molecule, despite the presence of polar covalent bonds within the molecule.

2.2.1 Define what a *covalent bond* is. (2)

2.2.2 Explain fully how hex-1-ene is a non-polar molecule, despite the presence of polar covalent bonds within the molecule. (4)

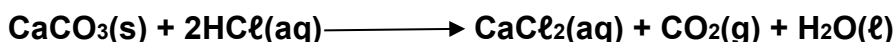
2.3 Intermolecular forces determine the phase of a substance.

2.3.1 Define *intermolecular force*. (2)

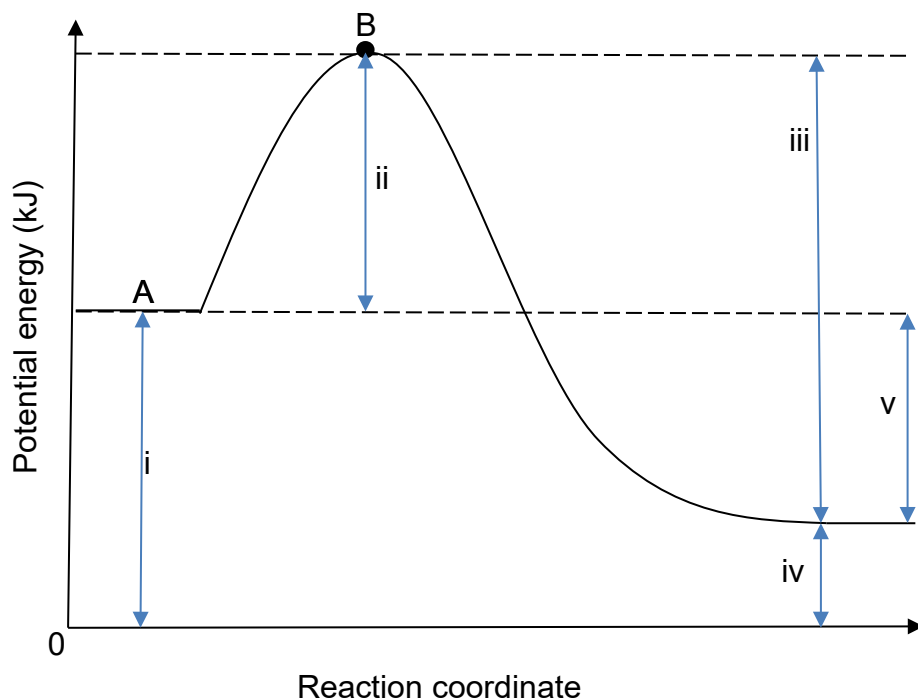
2.3.2 At room temperature, I_2 is a solid, whereas H_2O is a liquid. Explain this difference in phase, with reference to the intermolecular forces present in each substance. (5)

QUESTION 3 (IEB 2014)**17 MARKS**

In order to investigate the rate of the reaction between a carbonate and an acid, calcium carbonate and excess of $2 \text{ mol} \cdot \text{dm}^{-3}$ hydrochloric acid react in a reaction vessel. The balanced chemical equation for this reaction is



Consider the potential energy profile for this reaction as illustrated below:



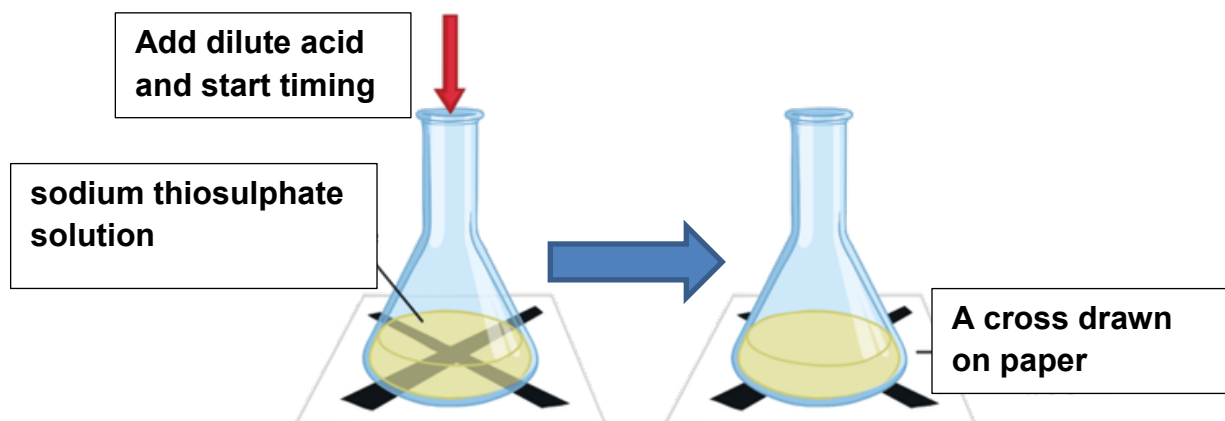
- 3.1 The graph has been labelled I – V to represent the various energies that are illustrated by this energy profile. Identify each of these energies on the profile. (5)
- 3.2 Using the molecular collision theory, explain why the chemical reaction must gain potential energy between position A and position B according to the energy profile. (2)
- 3.3 Provide a name for the position on the graph labelled B and what significant process takes place at this point. (2)
- 3.4 Name the type of reaction this graph represents. Explain how you came to this conclusion. (3)
- 3.5 The same reaction takes place in the presence of a catalyst. Sketch the potential energy profile and add a dotted-line to show how the graph would change in the presence of a catalyst. (2)
- 3.6 How would the presence of a catalyst affect the value of the energy labelled V? Explain your answer. (3)

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QUESTION 4

26 MARKS

Marcel wants to investigate the reaction between sodium thiosulphate solution and dilute hydrochloric acid.



Marcel recorded the time taken for the reaction mixture to become too cloudy to see the cross. He used different concentrations of sodium thiosulphate solution combined with a fixed volume of dilute hydrochloric acid. Marcel recorded data for six different concentrations of thiosulphate solution.

NOTE: All reactions are carried out in the same atmospheric conditions.

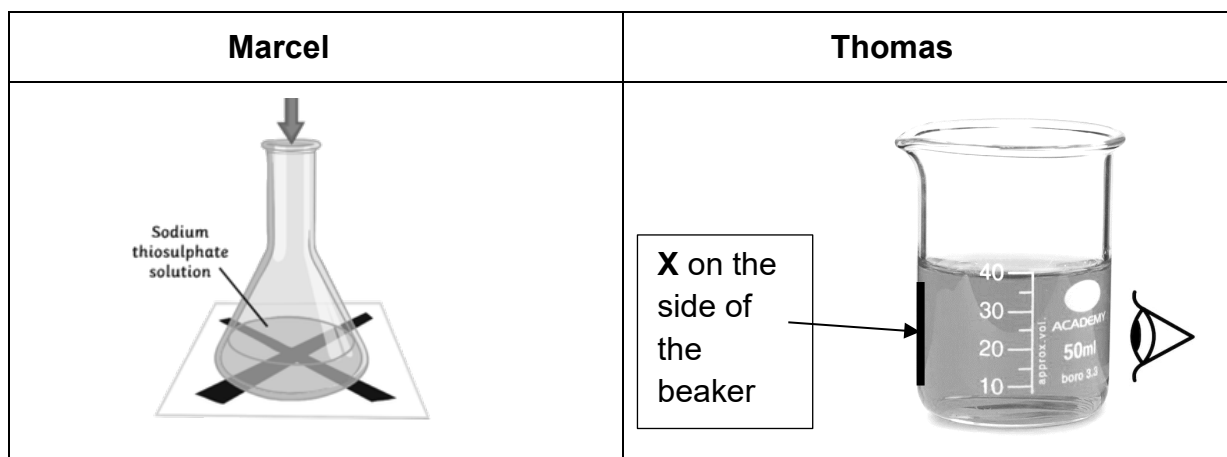
Marcel's results are shown in the table below:

Concentration of $\text{Na}_2\text{S}_2\text{O}_3$ solution ($\text{mol}\cdot\text{dm}^{-3}$)	Time taken for mixture to become too cloudy (s)				Reaction Rate ($\times 10^{-3} \text{ s}^{-1}$)
	Trial 1	Trial 2	Trial 3	Average	
0,10	126,2	124,7	125,4	125,4	7,97
0,15	81,4	83,8	84,5	83,2	12,02
0,25	49,2	49,3	49,6	49,4	20,24
0,40	31,3	31,4	31,2	31,3	31,95
0,45	27,2	27,9	28,4	27,8	35,97
0,50	23,5	24,8	26,7	Question 4.1.1	

4.1 Refer to the table of results to answer these questions.

- 4.1.1 Calculate and write down the missing average time for the solution with a concentration of $0,50 \text{ mol}\cdot\text{dm}^{-3}$. (2)
- 4.1.2 Which thiosulphate solution has the most precise results? (1)
- 4.1.3 Write down the equation Marcel used to calculate the Reaction Rate values. (2)

- 4.1.4 Complete the graph of concentration vs reaction rate on the Answer Sheet by...
- filling in the boxes to indicate the scale on the y-axis. (2)
 - plotting the first five relevant points from the table and drawing the line of best fit. (5)
- 4.1.5 Refer to the graph and the table of results to help you state the mathematical relationship between:
- the reaction rate and concentration of sodium thiosulphate solution for the solution to become too cloudy. (2)
 - the concentration of sodium thiosulphate solution and the time taken for the solution to become too cloudy. (2)
- 4.2 Marcel carries out exactly the same reactions on a colder day. The reaction rate values calculated are different.
- 4.2.1 Draw a line on the same graph paper as Question 4.1.5 for the results he would obtain on a colder day. Label this line "Cold". (2)
- 4.2.2 Explain, in terms of the collision theory, the effect of a decrease in temperature on reaction rate. (5)
- 4.3 Thomas and Marcel use the same reactions on a third day but they each used slightly different apparatus and methods as shown in the diagrams below.



Both students timed how long it would take for the mark to "disappear" at 20 °C for two different experiments.

Experiment 2 was the same as Experiment 1 except that the volumes of all solutions were doubled.

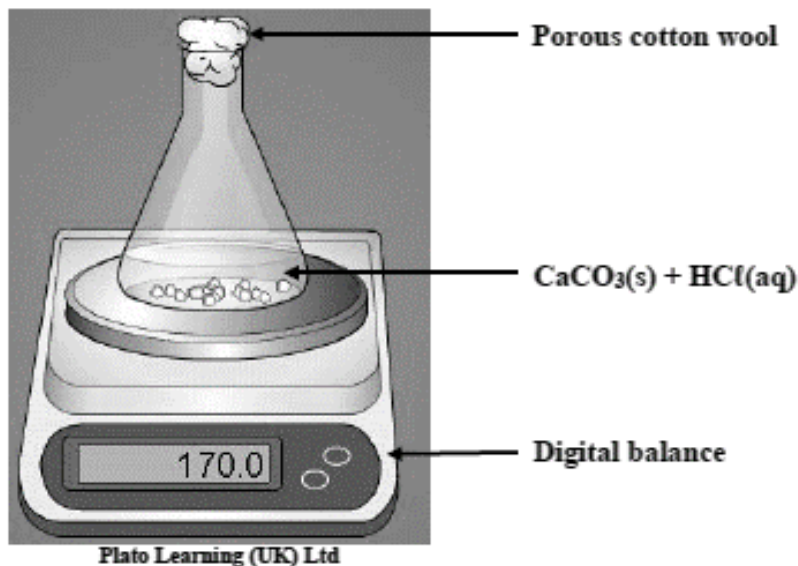
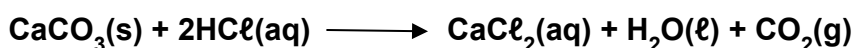
	Experiment 1 time (s)	Experiment 2 time (s)
Marcel	18	7
Thomas	10	Question 4.4.2

Given that they made NO errors in their experiments:

- 4.3.1 explain why Marcel's time for Experiment 2 was lower than it was for Experiment 1. (2)
- 4.3.2 write down the value for Thomas's Experiment 2. (2)

QUESTION 5 (IEB 2015)**18 MARKS**

Mandy and Siya conduct a series of experiments to investigate the effect of changing temperature, concentration and surface area on the rate of reaction between hydrochloric acid and calcium carbonate. The balanced chemical equation for the reaction taking place is:



They carry out each experiment in a conical flask, which is placed on top of a sensitive digital balance as shown in the diagram below. They monitor the decrease in mass of the conical flask and its contents over a period of time.

- 5.1 Mandy makes the following statement: 'The reason why the mass of the conical flask and its contents decreases over a period of time is because the reactants are being used up.' Do you agree with Mandy? Explain your answer. (2)

Mandy and Siya carry out FOUR different experiments using the conditions indicated in the table below.

Experiment	Temperature (°C)	Concentration of HCl (mol·dm ⁻³)	State of CaCO ₃
1	20	0,1	Large pieces
2	15	0,1	Large pieces
3	20	0,1	Powder
4	20	0,2	Large pieces

In each of their four experiments Mandy and Siya did the following:

- They used the same volume of hydrochloric acid.
- They used the same mass of calcium carbonate.
- They ensured the calcium carbonate was always in excess.
- They ensured the calcium carbonate was always fully covered by the acid.

- 5.2 Explain, in terms of the collision theory, the effect of a surface area on reaction rate. (4)

The results of EXPERIMENT 1 were used to sketch the graph shown on your ANSWER SHEET.

- 5.3 On the same set of axes used to sketch the graph for Experiment 1, on your ANSWER SHEET, draw sketch graphs to represent the results of Experiments 2, 3 and 4. Use of a SHARP, DARK pencil for each graph is recommended. CLEARLY LABEL EACH GRAPH using the key indicated below.

Experiment 2 Dashed line - - - - -

Experiment 3 Solid line _____

Experiment 4 Dash-dot line — . — . — . — . — . — . (6)

- 5.4 The Maxwell-Boltzmann curve below shows the distribution of molecular energies of reacting molecules in a chemical reaction at low temperature.

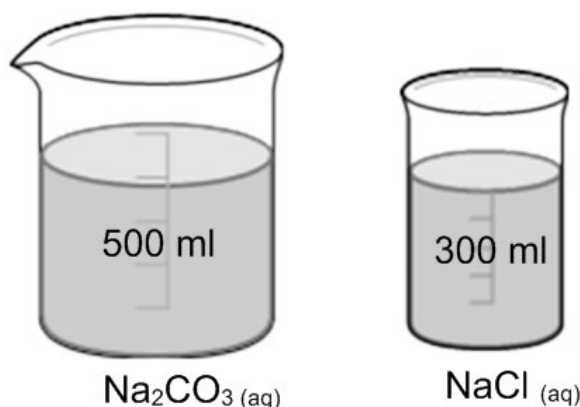
5.4.1 Define *activation energy*. (2)

5.4.2 State what is represented by the shaded area shown on the above graph. (2)

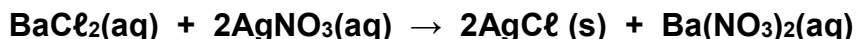
5.4.3 On the same set of axes used to draw the Maxwell-Boltzmann curve in Question 5.3, on your ANSWER SHEET, draw a new curve indicating the distribution of molecular energies at a **higher temperature**. Use of a SHARP, DARK pencil is recommended. (2)

QUESTION 6**33 MARKS**

- 6.1 A chemist makes a $0,5 \text{ mol.dm}^{-3}$ standard solution of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3 \text{ (aq)}$) using the usual equipment.
- 6.1.1 Define *concentration*. (2)
- 6.1.2 Calculate the mass of $\text{Na}_2\text{S}_2\text{O}_3$ needed to make 500 cm^3 of this standard solution. (4)
- 6.1.3 Should the solid $\text{Na}_2\text{S}_2\text{O}_3$ be added to a volumetric flask before or after distilled water is added? Briefly explain your answer. (2)
- 6.2 John and James have $761,6 \text{ dm}^3$ of ammonia, NH_3 and $1971,2 \text{ dm}^3$ of carbon dioxide, CO_2 at STP. John says that they have the same amount of substance, but James says they don't. Who is correct? Explain your answer. (4)
- 6.3 In the diagram right, the concentration of the Na_2CO_3 solution is $0,6 \text{ mol.dm}^{-3}$ while the concentration of the NaCl solution is $0,2 \text{ mol.dm}^{-3}$.



- 6.3.1 Write dissociation equations for both solutions. (4)
- 6.3.2 The two solutions are mixed. What is the concentration of Na^+ ions in the mixture? (4)
- 6.4 Barium chloride and silver nitrate react according to the following equation to produce silver chloride and barium nitrate.

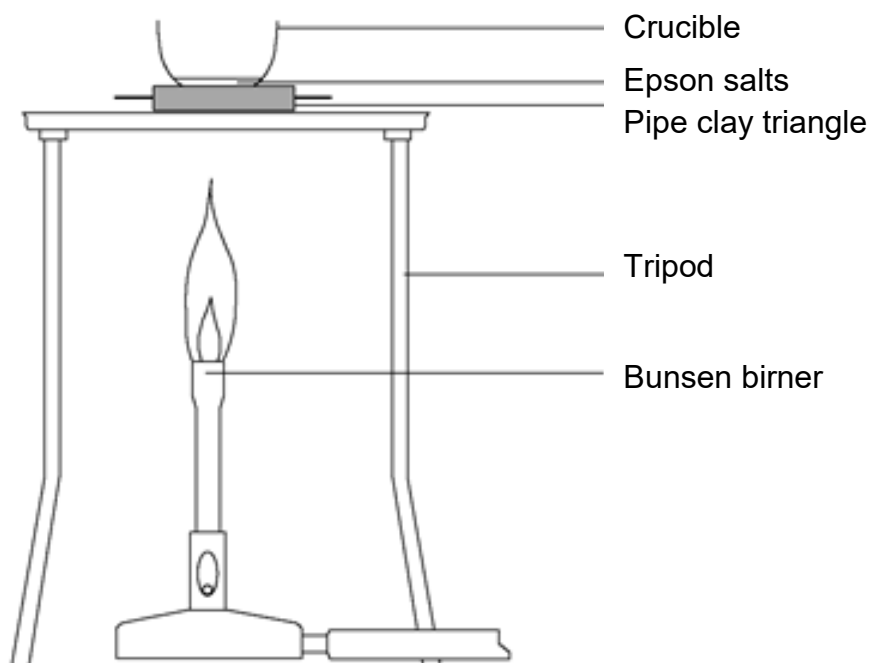


If you have $1,56 \text{ g}$ of BaCl_2 and $2,96 \text{ g}$ of AgNO_3 :

- 6.4.1 What mass of AgCl will you produce? (6)
- 6.4.2 Calculate the percentage yield if $1,82 \text{ g}$ of AgCl is actually collected after the reaction. (3)

- 6.5 In an experiment, the water of crystallisation is removed from hydrated Epsom salts to find x in the formula: $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$.

The apparatus below is used to gather the information in the table.



Complete the table by writing the question number in the block in your answer book.

(4)

(a) Mass of crucible	22,480	g
(b) Mass of crucible + Epsom salts	24,943	g
(c) Mass of crucible + Epsom salts - water	23,683	g
(d) Mass of Epsom salts	2,463	g
(e) Mass of Epsom salts - water	1,203	g
(f) Mass of water	1,260	g
(g) Moles of Epsom salts - water	6.5.1	mol
(h) Moles of water	6.5.2	mol
(i) ratio moles MgSO_4 : moles H_2O	6.5.3	
(j) Formula of Epsom salts	6.5.4	

QUESTION 7**26 MARKS**

Organic molecules undergo a range of chemical reactions based on their structure.

A number of molecules is listed in the table below.

X	$\begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} = & \text{C} - & \text{C} - & \text{C} - & \text{C} - & \text{H} \\ & & & & & & \\ & & & \text{H} & \text{H} & \text{H} & \end{array}$
Y	$\begin{array}{ccccccc} & & \text{H} & & \text{H} & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} = & \text{C} - & \text{C} = & \text{C} - & \text{C} - & \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array}$

- 7.1 Give the condensed formula of molecule X. (2)
- 7.2 Write down the IUPAC name of molecule Y. (3)
- 7.3 You have been given 2 colourless liquids. You are told that one is molecule X and that the other is molecule Y.
- Explain how you would perform a chemical test to determine which substance is molecule X and molecule Y. (3)
- 7.4 Molecule X is a product of the thermal cracking of octane.
- 7.4.1 Define saturated. (2)
- 7.4.2 Using molecular formulae, write a balanced chemical equation for the thermal cracking of octane to form molecule X and one other product. (4)
- 7.4.3 Which of the two products would have the higher boiling point? (1)
- 7.5 Using molecular formulae, write a balanced chemical equation for the complete combustion of molecule Y. (4)
- 7.6 Compound X is treated with hydrogen gas in the presence of a hot nickel catalyst.
- 7.6.1 Identify the TYPE of reaction that occurs. (1)
- 7.6.2 Name the homologous series of the organic product formed. (2)
- 7.6.3 What is the purpose of the nickel catalyst? (1)

QUESTION 8**17 MARKS**

Organic molecules undergo a range of chemical reactions based on their structure.

A number of molecules is listed in the table below.

A	$\begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3\text{CH}_2-\text{C}-\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
B	$\begin{array}{c} \text{O} \\ \\ \text{H}-\text{C} \\ \\ \text{O}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
C	$\begin{array}{ccccccc} & \text{H} & \text{O} & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \end{array}$
D	$\begin{array}{ccccc} & \text{H} & \text{H} & \text{O} & \\ & & & & \\ \text{H}- & \text{C} & -\text{C} & -\text{C} & \\ & & & & \\ & \text{H} & \text{H} & \text{O}-\text{H} & \end{array}$

8.1 Write down the IUPAC name of

8.1.1 Molecule A (3)

8.1.2 Molecule C (3)

8.3 Identify the homologous series of

8.2.1 Molecule B (1)

8.2.2 Molecule C (1)

8.2.3 Molecule D (1)

8.4 Molecule B and D are isomer.

8.3.1 Define the term *isomer*. (2)

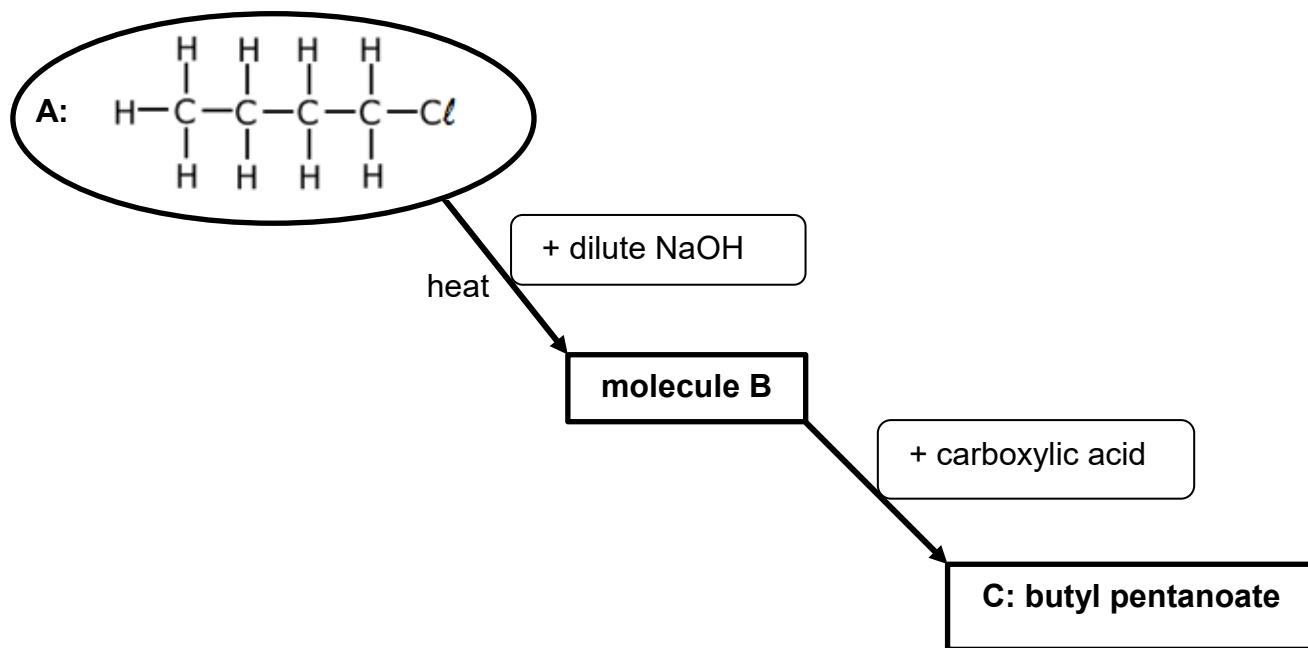
8.3.2 Name the type of isomer. (2)

8.3.3 Both isomers have three carbons, but very different boiling points.
Account for this difference with reference to the relevant intermolecular forces in each of these molecules. (4)

QUESTION 9

22 MARKS

Consider the reaction scheme below:



- 9.1 Define homologous series. (2)
- 9.2 Give the homologous series for
- 9.2.1 molecule A (1)
- 9.2.2 molecule C (1)
- 9.3 Consider the reaction for molecule A to change to molecule B.
- 9.3.1 Name the specific type of substitution reaction. (1)
- 9.3.2 Give the molecular formula of molecule B. (2)
- 9.3.3 A student tries to make molecule B from molecule A, but uses concentrated NaOH and forms different products.
- (a) Name the specific type of reaction the now occurred. (2)
- (b) Name the inorganic product that formed. (1)
- 9.4 Molecule B reacts with a carboxylic acid to form butyl pentanoate.
- 9.4.1 Name the carboxylic acid used. (2)
- 9.4.2 Using **structural** formulae, write a balanced chemical equation to represent this reaction. Include all reaction conditions. (8)

END OF EXAMINATION